

Session 5.3: Clinical validation/calibration of MAVEs

Session Handout

Useful Links

- ClinVar: <https://www.ncbi.nlm.nih.gov/clinvar/>
- ACMG v3.0 Framework <https://doi.org/10.1038/gim.2015.30>
- Tavtigian et al, 2018 (ACMG framework as a Bayesian classifier)
<https://doi.org/10.1038/gim.2017.210>
- Brnich et al, 2019 (Validating functional assay data for variant classification)
<https://doi.org/10.1186/s13073-019-0690-2>

Glossary of useful terms

Term	Definition
ACGS	The Association for Clinical Genomic Science: UK professional association for scientists working within genetics. Developed UK-specific guidelines for variant classification.
ACMG	American College of Medical Genetics and Genomics: American professional association for medical genetics teams. Developed standards and guidelines for variant classification in 2015)
Benign	A variant which does not cause the disease of interest
BS3	Evidence code describing functional data in the 2015 ACMG v3.0 guidelines (BS3 = functional data used as evidence towards benignity)
ClinGen	National Institutes of Health (NIH)-funded resource dedicated to building a central resource that defines the clinical relevance of genes and variants for use in precision medicine and research.
ClinVar	Archive storing previously submitted variant classifications
EPs	Evidence (or exponent) points, a quantification of the amount of evidence afforded by individual evidence items, calculated as $\log_{2.08}$ of the OddsPath. Can be summed to calculate the overall evidence strength towards pathogenicity/benignity.
Evidence	Information and data used in a variant classification. In the ACMG v3.0 guidelines, evidence is assigned categorical evidence strengths towards either pathogenicity or benignity (on a scale: very strong, strong, moderate, or supporting)
“Hard” conflict	Conflicting interpretation of a variant arising when it has independent classifications of both (likely) pathogenic and (likely) benign
LLR	Log Likelihood Ratio
OddsPath	Odds of Pathogenicity, the likelihood of the evidence if the variant is truly pathogenic
Pathogenic	A variant which causes the disease of interest
Prior Probability	The starting assumption of a variant being disease-causing, set at 10% for the SVI Tavtigian framework
Posterior Probability	The probability of the variant causing disease given both our starting assumption and other evidence input into Bayes’ Rule
PS3	Evidence code describing functional data in the 2015 ACMG v3.0 guidelines (PS3 = functional data used as evidence towards pathogenicity)
“Soft” conflict	Conflicting interpretation of a variant arising when it has independent classifications of both (likely) pathogenic and uncertain significance, or (likely) benign and uncertain significance
SVC	ClinGen Sequence Variant Classification working group (previously the Sequence Variant Interpretation (SVI) group). Produce expert guidelines to support the ACMG general guidelines. https://www.clinicalgenome.org/tools/clingen-variant-classification-guidance/
SVC Brnich	Specific methodology for application of functional evidence in variant classification, produced by the ClinGen SVC working group
SVC Tavtigian	Refers to two publications produced by the ClinGen SVC working group that transition the categorical evidence strengths used by the ACMG v3.0 guidelines to a numerical points-based system.
Truthset	A reference set of previously classified variants which is used to assess how well an assay discriminates between pathogenic and benign variants.
VCEP	Variant Curation Expert Panel, a group of experts focussing on a single gene or set of related genes, and responsible for developing gene-specific variant interpretation guidelines

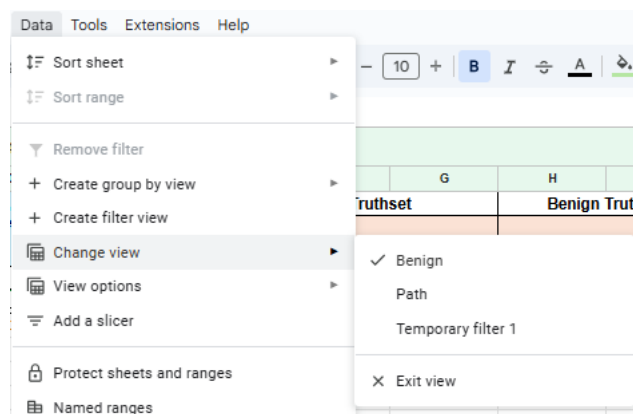
Practice session 1: Defining variant truthsets

Please open the file named 'session5.3_practicesession_part1' within Google Sheets (link: https://docs.google.com/spreadsheets/d/19w5E7MTQ99KI045VY_bt4yJCPDbiEhn2jaaNBUKG7-Y/edit?usp=sharing) **and make a copy within your own Google Sheets files (File → Make a copy).**

For your assigned gene, either *VHL* or *BRCA1*, generate a set of six possible truthsets for MAVE validation by performing the following filters for each respective tab. The columns to use for filtering in this part are highlighted in green.

Record on each tab the number of variants in your Pathogenic and Benign truthsets (as defined below).

Suggestion: To make the second practice session easier, you should save the filtered views for your pathogenic and benign truthsets separately using the 'Data' tab (Data → Save filter view (once filtering complete), Data → Change view → Exit view (to avoid further filters being applied to your kept view)



These filtered views can then be readily accessed again in the second practice session by selecting them from the drop-down menu (see screenshot above)

Truthset 1: ClinVar P/LP and B/LB ($\geq 1^*$), no conflicting interpretations

Both truthsets

- ≥ 1 star classifications (field: clinvar_stars)
- All variant types (field: Consequence)

Pathogenic truthset

- Pathogenic and/or Likely Pathogenic in ClinVar (field: clinvar_classification)
 - This includes "Likely pathogenic/Pathogenic"

Benign truthsets

- Benign and/or Likely Benign in ClinVar (field: clinvar_classification)
 - This includes "Likely benign/Benign"

Truthset 2: ClinVar P/LP and B/LB ($\geq 2^*$), no conflicting interpretations

Both truthsets

- ≥ 2 star classifications (field: clinvar_stars)
- All variant types (field: Consequence)

Pathogenic truthset

- Pathogenic and/or Likely Pathogenic in ClinVar (field: clinvar_classification)

Benign truthset

- Benign and/or Likely Benign in ClinVar (field: clinvar_classification)

Truthset 3: ClinVar P/LP and B/LB ($\geq 1^*$), including soft-conflicting interpretations

Both truthsets

- ≥ 1 star and 'conflicting' classifications (field: clinvar_stars)
- All variant types (field: Consequence)

Pathogenic truthset

- Pathogenic and/or Likely Pathogenic in ClinVar, including those also classified as Uncertain Significance (field: clinvar_classification)
 - *Hint: Do not include variants with both (likely) pathogenic and (likely) benign classifications in ClinVar*

Benign truthset

- Benign and/or Likely Benign in ClinVar, including those also classified as Uncertain Significance (field: clinvar_classification)

Truthset 4: ClinVar P/LP and B/LB ($\geq 1^*$), missense only, no conflicting interpretations

Both truthsets

- ≥ 1 star classifications (field: clinvar_stars)
- Variant type, or "consequence", containing "missense_variant" (field: Consequence)

Pathogenic truthset

- Pathogenic and/or Likely Pathogenic in ClinVar (field: clinvar_classification)

Benign truthset

- Benign and/or Likely Benign in ClinVar (field: clinvar_classification)

Truthset 5: ClinVar missense P/LP ($\geq 1^*$), systematically identified benign missense variants

Pathogenic truthset

- ≥ 1 star classifications (field: clinvar_stars)
- Variant consequence containing "missense_variant" (field: Consequence)
- Pathogenic and/or Likely Pathogenic in ClinVar (field: clinvar_classification)

Benign truthset

- Systematically identified benign variants (field: systematic_benign_variant)

Truthset 6: All nonsense and synonymous variants

Pathogenic truthset

- Variant consequence containing "stop_gained" (the name given to nonsense variants by VEP; field: Consequence)

Benign truthset

- Variant consequence containing "synonymous_variant" (field: Consequence)

Practice session 2: Assessing evidence weighting

Please open the file named 'session5.3_practicesession_part2'

(<https://docs.google.com/spreadsheets/d/1aUJTWwB7uETnkn-qMpzB0NIrqvfOt2mzEiWg-GeGRLY/edit?gid=0#gid=0>) **and make a copy within your own Google Sheets files (File → Make a copy).**

In this section, you will use the Brnich framework to quantify the evidence strength applicable for a MAVE in your assigned gene when using the different truthsets you have constructed. Using your six pre-filtered truthsets from the first practice session, assess the evidence weighting which should be applied for each assay.

To do this, enter into the Part 2 worksheet the number of variants in your pathogenic and benign truthsets which have a functionally abnormal, functionally normal, or intermediate result (see example below):

Truthset Name	Truthset Content	# Classified			Proportion pathogenic (Prior, P1)	Assay result						"+1 proportions" (Posterior, P2)		OddsPath=[P2 x (1- P1)] / [(1- P2) x P1]	
		Total	Pathogenic truthset	Benign truthset		Pathogenic truthset			Benign truthset			Pathogenic	Benign	Pathogenic	Benign
						Functionally abnormal	Intermediate	Functionally normal	Functionally abnormal	Intermediate	Functionally normal				
EXAMPLE	EXAMPLE	17	10	7	0.59	10	0	0	1	0	6	0.833	0.143	3.475	0.116

Hint: Use the Function_Class columns to filter your pathogenic and benign truthsets based on assay output (Abnormal, Intermediate, or Normal)